Compactly Arranging Every Way of Filling an L-Shaped Grid into a Non-Repeating Array

Rachel Rockey

Departmental Affiliation: Mathematics and Computer Science
College of Arts and Sciences

A de Bruijn Array (also called a torus) is a toroidal array of numbers where each filling of an *m*-by-*n* matrix, with digits chosen from 0 to k-1, is present only once. While it is well understood how to find a de Bruijn Array for fillings of an *m*-by-*m* rectangle (Jackson, Stevens, Hurlbert, 2009), arrays for other shapes are unstudied. I have worked to answer the question: can de Bruijn Arrays be found with different shapes of fillings? In particular, I have considered The L Problem. Instead of arranging fillings of a rectangular grid, this problem arranges fillings of a 2-by-2 grid with one square removed. I have proven that a de Bruijn Array does exist for every alphabet using this pattern.

Information about the Author:
Rachel Rockey is a sophomore secondary mathematics education major and a student in Christ College. She is a member of the MSEED Program, which helped to fund her research under NSF grant number 1068346. She first became interested in this topic after hearing Adam Goyt's talk on de Bruijn sequences in a math colloquium. Rachel hopes to one day teach middle school or high school mathematics.

Faculty Sponsor: Dr. Lara Pudwell

Student Contact: rachel.rockey@valpo.edu